ECO 2013 — Four-Week Macroeconomics Study Plan

Dhanesh | UCF Summer 2026

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Contents

ECO 2013 — Four-Week Macroeconomics Study Plan

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Course: ECO 2013 Principles of Macroeconomics, UCF Summer 2026
Instructors: Dr. Jordan Izenwasser & Dr. Nora Underwood
Textbook: Karlan & Morduch, *Macroeconomics* 2024 ed. (McGraw-Hill, via Connect)
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> [i] **AI Use in ECO 2013:** AI tools are permitted as a supplement — use Claude or Khan Academy to understand *why* answers are correct. Do not use AI to complete Connect assignments for you; the Smartbook and homework exist to build the understanding you need for in-person exams.

How to Use This Plan

Each week follows the four-step study sequence:

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Step 1 → Textbook (Karlan & Morduch chapters via Connect)
Step 2 → Online resources (Khan Academy, Lumen, MRU)
Step 3 → Write your own notes (key concepts + worked numbers)
Step 4 → Practice questions and Smartbook/Connect prep
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Exam Calendar (keep visible)

- **Midterm 1** — covers Chapters 1–3 (Weeks 1–4 of semester)
- **Midterm 2** — covers Chapters 7–10 (Weeks 5–8 of semester)
- **Final** — cumulative; 60 questions, 150 minutes
- Exams are in-person at the College of Business testing center; bring TI-30Xa or TI-30X IIS calculator

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Week 1 — Economics and Life + Specialization and Exchange (Chapters 1–2)

What these chapters are about: Chapter 1 establishes the core logic of all of economics: resources are scarce, so every choice has a cost. Chapter 2 shows why trade makes everyone better off even when one party is worse at everything — the key is *comparative* advantage, not absolute.

Cross-domain connection (for Dhanesh): Opportunity cost is like impedance in a circuit. When current can flow two ways, it takes the path of least resistance — not because it has "decided" to, but because that is where the cost is lowest. Comparative advantage is the same idea: specialise where your opportunity cost is lowest, then exchange. Impedance matching in RF design is this principle applied to power transfer.

Day 1 — Textbook (Step 1): Chapter 1

Read: Chapter 1 — *Economics and Life* (via Connect)

Do the Smartbook for Chapter 1. It will adapt to what you know and keep going until you've mastered the definitions. Do not skip this — the Smartbook questions are close to what appears on exams.

Sit with these questions while reading: - The textbook says "all decisions happen in an environment of scarcity." Can you name three scarcity trade-offs you personally faced this week — not just money, but time and attention? - The marginal principle says ignore sunk costs and compare only what changes. Why is this hard to do in practice? - What is the difference between efficiency and equity? Can a policy be efficient but unfair?

Day 2 — Textbook (Step 1): Chapter 2

Read: Chapter 2 — *Specialization and Exchange* (via Connect)

Do the Smartbook for Chapter 2.

Key calculation to practise: The Production Possibilities Frontier (PPF) and calculating opportunity costs. Work through at least one numerical example where you compute the opportunity cost of producing one more unit of Good A in terms of Good B — then reverse it. Make sure you get both directions right.

Sit with these questions: - Country A can produce either 100 shirts or 50 computers. Country B can produce either 60 shirts or 60 computers. Which country has a comparative advantage in computers? Work it out using opportunity cost before reading the answer. - Why do both countries gain from trade even if one country is worse at producing everything?

Day 3 — Online Resources (Step 2)

Work through: - **Lumen Learning — Comparative Advantage and Gains from Trade** — Read the Saudi Arabia/US trade example; then try to replicate the table yourself on paper - **OpenStax — Absolute and Comparative Advantage** — Second treatment of the same concept; if one explanation didn't click, this one might - **Khan Academy — Basic Economic Concepts (Macroeconomics)** — Do the exercises on scarcity and PPC - **Pearson — PPF, Comparative Advantage and Trade (video)** — Watch the video; then try drawing your own PPF and labelling it

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Day 4 — Notes + Practice (Steps 3 & 4)

Write your own notes covering: - What scarcity is and why it forces trade-offs (your own words; do not copy definitions) - The marginal principle — one example you made up yourself - How to calculate opportunity cost from a PPF table (write out the steps) - The difference between absolute and comparative advantage (a diagram helps) - Why trade makes both parties better off (in your own example)

Practice quiz sites: - **Quizlet — ECO 2013 Exam 1 Flashcards** — Run through all cards; mark anything you hesitate on - **Quizlet — Karlan/Morduch Chapter 1 Key Terms** — These are from your exact textbook - **Quizlet — ECO2013 Macroeconomics Exam 1** — Broader exam set; covers Chapters 1–5

Self-check questions (answer before looking): 1. Dhanesh has 4 hours. He can spend them studying for ECO (gaining 8 exam points) or working a shift (earning $40). What is the opportunity cost of studying? What is the opportunity cost of working? 2. Country X can produce 200 tons of wheat OR 100 tons of steel. Country Y can produce 150 tons of wheat OR 150 tons of steel. Who has a comparative advantage in wheat? Who in steel? 3. What does it mean that the PPC bows outward (is concave)? What does a straight-line PPC mean? 4. Name two things that shift the PPC outward and explain why. 5. Define marginal benefit and marginal cost. Why does rational decision-making focus on the margin?

Week 1 Key Terms

Term	Meaning
Scarcity	Unlimited wants, limited resources — forces all choices
Opportunity cost	The value of the best alternative you gave up
Marginal analysis	Comparing extra cost vs. extra benefit of one more unit
Trade-off	Giving up some of X to get more of Y
PPC / PPF	Graph showing maximum combinations of two goods an economy can produce
Absolute advantage	Producing more output with the same resources

Term	Meaning
Comparative advantage	Lower opportunity cost of production — the real basis for trade
Gains from trade	Extra total output created when each party specialises in their comparative advantage

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Week 2 — Markets: Supply and Demand (Chapter 3)

What this chapter is about: How prices are set. Supply and demand is the core model of economics. Get this chapter right — almost every later topic (GDP, monetary policy, labour markets) uses it as a foundation.

Cross-domain connection: Supply and demand equilibrium is a feedback control system. The price is the signal. When price is above equilibrium, surplus causes it to fall; when below, shortage causes it to rise. This is negative feedback — a self-correcting loop. You'll recognise it from control systems: the market is the plant, price is the error signal, and buyers and sellers are the feedback path.

Day 1 — Textbook (Step 1): Chapter 3, Part 1 (Demand)

Read: Chapter 3, demand side (via Connect). Do the Smartbook for the demand section.

Draw, don't just read: Draw a demand curve. Label price on the Y-axis, quantity on the X-axis. Then practice shifting it — draw what happens when: - Consumer income rises (for a normal good) - The price of a complement falls - Future prices are expected to rise

Sit with these questions: - The law of demand says price and quantity demanded move in opposite directions. Why? (Think: what happens to your willingness to buy something as it gets more expensive?) - What is the difference between a *change in quantity demanded* (movement along the curve) and a *change in demand* (shift of the curve)? This distinction comes up on every exam.

Day 2 — Textbook (Step 1): Chapter 3, Part 2 (Supply + Equilibrium)

Read: Chapter 3, supply and equilibrium sections (via Connect). Do the Smartbook for these sections.

Draw, don't just read: Draw supply and demand on the same graph. Mark equilibrium. Then run four scenarios: 1. Demand increases → what happens to price and quantity? 2. Supply decreases → what happens? 3. Both demand increases and supply decreases → what happens to price? What about quantity (hint: it depends)? 4. A price ceiling is set below equilibrium → what happens?

Check your graphs make sense economically before moving on.

Day 3 — Online Resources (Step 2)

Work through: - **Khan Academy — Supply, Demand, and Market Equilibrium** — Do every exercise in this unit. This is the most important practice you can do for the midterm. - **Khan Academy — Changes in Equilibrium (video)** — Watch this and pause to draw the graph yourself before they show the answer - **Khan Academy — Market Equilibrium Exercise (with price controls)** — Practice price floors and ceilings

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Day 4 — Notes + Practice (Steps 3 & 4)

Write your own notes covering: - The five demand shifters (income, prices of related goods, tastes, expectations, number of buyers) - The five supply shifters (input costs, technology, prices of related goods in production, expectations, number of sellers) - How to read a supply and demand graph and identify equilibrium - What happens to equilibrium P and Q for each of the four shift scenarios you practised

Practice quiz sites: - **Khan Academy Supply/Demand exercises** — Complete all exercises - **Quizlet — ECO 2013 Exam 1 Flashcards** — Focus on Chapter 3 content - **Quizlet — Module 1 Foundations of Economics (Chapters 1–4)**

Self-check questions: 1. What happens to equilibrium price and quantity when demand falls and supply also falls? (Work it out on paper — draw the graphs.) 2. The government sets a price ceiling on petrol below the equilibrium price. Draw the result. Is there a surplus or shortage? By how much does quantity demanded exceed quantity supplied? 3. Name two things that would shift the supply of coffee to the left (decrease supply). 4. Is a change in the price of good X itself a "supply shifter" or a "demand shifter"? Explain why. 5. What is the difference between a price floor and a price ceiling? Give one real-world example of each.

Week 2 Key Terms

Term	Meaning
Law of demand	Price ↑ → quantity demanded ↓ (inverse relationship)
Law of supply	Price ↑ → quantity supplied ↑ (positive relationship)
Equilibrium	Price where quantity demanded = quantity supplied
Surplus	Quantity supplied > quantity demanded (price above equilibrium)
Shortage	Quantity demanded > quantity supplied (price below equilibrium)
Demand shifters	Factors that move the whole demand curve (income, related prices, etc.)
Supply shifters	Factors that move the whole supply curve (input costs, technology, etc.)
Price ceiling	Legal maximum price (causes shortage if below equilibrium)
Price floor	Legal minimum price (causes surplus if above equilibrium)

Week 3 — Measuring GDP + The Cost of Living (Chapters 7–8)

What these chapters are about: How do we measure how well an entire economy is doing? Chapter 7 introduces GDP — the scoreboard. Chapter 8 introduces inflation — the adjustment that lets you compare scores across time. Together they are the foundation of macroeconomic measurement.

Cross-domain connection: GDP is like measuring total power output of a system. Real GDP is power output corrected for voltage drift (inflation). If your signal grows but your supply voltage also rose, the actual performance didn't improve as much as the raw reading suggests. Nominal GDP is the uncorrected reading; real GDP removes the "price inflation noise" to show what actually happened to output.

Day 1 — Textbook (Step 1): Chapter 7

Read: Chapter 7 — *Measuring GDP* (via Connect). Do the Smartbook for Chapter 7.

The formula you must memorise:

> **GDP = C + I + G + NX** - C = Consumer spending (households) - I = Investment (businesses buying capital) - G = Government purchases (not transfers) - NX = Net exports (exports minus imports)

Practise with a number: Suppose C = $10 trillion, I = $2 trillion, G = $3 trillion, exports = $1.5 trillion, imports = $2 trillion. Calculate GDP. (Answer: $14.5 trillion)

Sit with these questions: - Why don't we count intermediate goods in GDP? (Hint: think about avoiding double-counting) - Why don't we count transfer payments (like Social Security) in G? - If someone builds and sells a house, how does that appear in GDP?

Day 2 — Textbook (Step 1): Chapter 8

Read: Chapter 8 — *The Cost of Living* (via Connect). Do the Smartbook for Chapter 8.

The formula you must memorise:

> **CPI = (Cost of basket in current year / Cost of basket in base year) × 100**
>
> **Inflation rate = [(CPI this year − CPI last year) / CPI last year] × 100**

Practise with a number: Base year basket costs \$200. This year it costs \$230. What is the CPI? What is the inflation rate from last year if last year's CPI was 110 and this year's is 115?

Sit with these questions: - What is "substitution bias" and why does it cause the CPI to overstate inflation? - What is the difference between the CPI and the GDP deflator? Which covers more goods? - If your salary rises by 3% but inflation is 5%, are you better or worse off in real terms?

Day 3 — Online Resources (Step 2)

Work through: - **Khan Academy — Parsing GDP (video)** — Watch this; pause after each component and write down an example of your own - **Khan Academy — Examples of Accounting for GDP (video)** — Work the numerical examples before they reveal answers - **Khan Academy — Limitations of GDP (video)** — Important for short-answer questions - **Fiveable — CPI Formula and Inflation** — Work through the CPI calculation examples

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Day 4 — Notes + Practice (Steps 3 & 4)

Write your own notes covering: - The four components of GDP (C + I + G + NX) with one real-world example of each - Why intermediate goods are excluded from GDP - The CPI formula and how to calculate inflation rate from two CPI values - Real vs. nominal GDP — what the GDP deflator does - Two limitations of GDP as a measure of wellbeing

Practice quiz sites: - **Khan Academy GDP exercises** — Complete all exercises - **Quizlet — GDP, Inflation, Unemployment & CPI Formulas** — Formula-focused; work through every card - **Quizlet — CPI, GDP, and Unemployment Flashcards** — Combined set for Chapters 7–9 - **Quizlet — GDP and CPI Practice Test** — Test format; timed practice

Self-check questions: 1. Government spends \$500 billion on military equipment. Does this count in C, I, G, or NX? 2. A US company manufactures steel in China and ships it back to sell in the US. How does this affect US GDP through NX? 3. Base year: basket costs \$500. Year 2: same basket costs \$550. What is the CPI in Year 2? 4. If nominal GDP grew 6% and the GDP deflator rose 4%, what was real GDP growth? 5. Name three things GDP doesn't measure that affect people's actual wellbeing.

Week 3 Key Terms

Term	Meaning
GDP	Total market value of all final goods and services produced in a country in a year
Expenditure approach	GDP = C + I + G + NX
Nominal GDP	GDP measured in current prices (not adjusted for inflation)
Real GDP	GDP adjusted for inflation (constant prices)
GDP deflator	Price index used to convert nominal to real GDP
CPI	Tracks price changes of a fixed basket of consumer goods
Inflation rate	% change in the price level from one year to the next
Substitution bias	CPI overstates inflation because it ignores consumers switching to cheaper goods

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Week 4 — Unemployment and the Labor Market + Economic Growth (Chapters 9–10)

What these chapters are about: Chapter 9 maps out the types of unemployment and what a "healthy" level looks like. Chapter 10 explains what makes an economy grow over the long run — not just short-term fluctuations but the sustained increase in living standards over decades.

Cross-domain connection: The three types of unemployment map cleanly onto signal theory. Frictional unemployment is like propagation delay — a small, expected lag between states. Structural unemployment is like a filter mismatch — the signal (worker skills) no longer matches the system (employer needs). Cyclical unemployment is like a noise burst — it comes and goes with the business cycle.

The "natural rate" is the system's baseline noise floor; you can't eliminate it without changing the fundamental physics.

Day 1 — Textbook (Step 1): Chapter 9

Read: Chapter 9 — *Unemployment and the Labor Market* (via Connect). Do the Smartbook for Chapter 9.

The classification you must know cold: - **Frictional unemployment** — temporarily between jobs; normal and healthy (e.g., a graduate searching for their first job) - **Structural unemployment** — skills mismatch; job seeker's skills no longer match what employers need (e.g., a coal miner in a region moving to renewables) - **Cyclical unemployment** — caused by a recession; falls with economic recovery (e.g., construction workers laid off during a housing crash) - **Natural Rate of Unemployment (NRU)** = frictional + structural; this is the "floor" even in a healthy economy

Sit with these questions: - A car factory automates its assembly line and lays off 200 workers. Is this frictional, structural, or cyclical? - An economy has 3% frictional and 2% structural unemployment. What is the NRU? If actual unemployment is 7%, what is the cyclical rate? - Why is zero unemployment not a realistic or even desirable goal?

Day 2 — Textbook (Step 1): Chapter 10

Read: Chapter 10 — *Economic Growth* (via Connect). Do the Smartbook for Chapter 10.

The Rule of 70 — memorise this:

> **Years to double = 70 / annual growth rate (%)**

If GDP grows at 2% per year, it doubles in 35 years. At 7%, it doubles in 10 years. Small differences in growth rates matter enormously over time.

Three drivers of long-run growth (know all three): 1. **Physical capital accumulation** — more machines, infrastructure, tools 2. **Human capital accumulation** — better-educated, healthier workforce 3. **Technological progress** — doing more with the same inputs

Sit with these questions: - Two countries have the same GDP today. Country A grows at 1% per year; Country B at 4%. How much larger will Country B's GDP be relative to Country A's after 35 years? (Use the Rule of 70 — don't need exact math, estimate the doubling pattern.) - Why does saving matter for economic growth? (Hint: think about where investment comes from.) - What role does education play in growth? Is it physical capital or human capital?

Day 3 — Online Resources (Step 2)

Work through: - **Khan Academy — Unemployment unit** — Videos and exercises; do all exercises - **Study.com — Inflation & Unemployment Relationship (Phillips Curve)** — Read the section on the Phillips Curve; it connects Chapters 8 and 9 - **Lumen Learning — Economic Growth chapter** — Good supplement to Karlan on growth drivers - **CORE Economics — Growth and Fluctuations** — Optional deeper reading on business cycles and long-run growth

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Day 4 — Notes + Practice (Steps 3 & 4)

Write your own notes covering: - The three types of unemployment with one original example of each - How to calculate the NRU and cyclical unemployment from given numbers - The Rule of 70 — two worked examples you made up - The three drivers of long-run growth with explanation of each - The Phillips Curve — what it shows and why it's controversial in the long run

Practice quiz sites: - **Quizlet — GDP, Unemployment, and Inflation Practice Quiz** — Test format; covers all three big measures together - **Quizlet — Macroeconomics Quiz: Unemployment and Inflation** — Chapter-level questions with specific examples - **Quizlet — Macroeconomics Inflation and Unemployment Practice** — Practice classifying unemployment types - **Khan Academy — Unemployment exercises**

Self-check questions: 1. A software developer quits her job to look for a better one. What type of unemployment? 2. A steelworker loses his job when the steel mill is replaced by an automated plant. What type? 3. During a recession, a restaurant lets go of half its staff. What type? 4. If frictional unemployment = 2% and structural = 3%, what is the NRU? If actual unemployment = 8%, what is cyclical unemployment? 5. Using the Rule of 70: if an economy grows at 3.5% per year, how many years until GDP doubles? 6. Which of the three growth drivers (physical capital, human capital, technology) do you think has the biggest impact on long-run growth? Prepare an argument for and against your choice.

Week 4 Key Terms

Term	Meaning
Frictional unemployment	Short-term job searching; healthy and expected
Structural unemployment	Skills mismatch; requires retraining to resolve
Cyclical unemployment	Caused by recessions; disappears with recovery
Natural Rate of Unemployment (NRU)	Frictional + structural; the economy's baseline
Labor force participation rate	% of working-age population that is employed or actively seeking work
Discouraged workers	People who stopped looking for work; not counted as unemployed
Rule of 70	Years to double = 70 / growth rate (%)
Human capital	Education, skills, and health of the workforce
Technological progress	Doing more output with the same inputs
Phillips Curve	Shows the short-run trade-off between inflation and unemployment

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Weeks 5–8 Preview (Post-Midterm 2 Topics)

After Midterm 2, the course moves into:

Topic	Chapter	Key concepts to preview
Aggregate Expenditure	11	AE = C + I + G + NX; multiplier; MPC
Economic Fluctuations	11 + Module	Business cycles; AD-AS model

Topic	Chapter	Key concepts to preview
The Basics of Finance	13	Present value; interest rates; bonds vs. stocks
Money and the Monetary System	15	M1/M2; fractional reserve banking; money multiplier; Fed tools

Best free resources to preview these: - **MRU — Monetary Policy Practice Questions** — for Chapter 15 - **EconEdLink — Money, Monetary Policy, and the Federal Reserve** — video series + quiz - **Quizlet — Money Creation & Federal Reserve**

General Resources (Use Every Week)

Resource	Use It For
Khan Academy — Macroeconomics	Videos + exercises for every topic in the course
Marginal Revolution University	Short, sharp economics videos — great alternative angle
Lumen Learning — Macroeconomics	Free open textbook; same topics as Karlan
OpenStax Macroeconomics 2e	Free peer-reviewed textbook; deeper than Karlan
Investopedia	Quick reliable definitions for any term
Quizlet — search “ECO 2013”	UCF-specific flashcard sets from past students
BLS.gov	Real CPI and unemployment data from the Bureau of Labor Statistics
BEA.gov	Real GDP data from the Bureau of Economic Analysis

Session End Checklist

After every study session:

- ☐ Did I complete the Smartbook for this chapter? (It runs until mastery — don't stop early)
- ☐ Did I work at least one numerical problem by hand — not just read an explanation?
- ☐ Did I write notes in my own words — not copy-pasted from any source?
- ☐ Did I test myself on the self-check questions before looking at answers?
- ☐ What is the one concept I am least sure of — and where will I find a different explanation next session?

Exam reminder: Bring a TI-30Xa or TI-30X IIS calculator. No notes allowed. Calculator is available on the computer in the testing center as a backup.

Built for Dhanesh's personal study. AI tools may be used as a supplement in ECO 2013 per the syllabus AI policy. All graded assignments must be completed independently.